Rent or Toxicity?

Strategic exploitability of learning market makers with public wallet information

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6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

2026: the year retail leverage went wrong at national scale

−44%

KOSPI, June 19 to July 29

1.2 million

retail accounts margin-called:

3.4% of all adults

$39 billion

retail losses on single-stock leveraged ETFs

“Retail” means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- trading on a phone app, with borrowed money, against professionals they cannot see

President Lee Jae-myung, on X, October 14, 2026

살아야지요. 어떻게든 사셔야 합니다. 힘드시면 109번 전화를 하십시오.

사실 길, 살아야 하는 이유를 함께 찾아드릴 것입니다.

“You must live. Somehow, you must live. If it is hard, call 109. We will find, together, a way and a reason to live.”

Sources: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, “KOSPI Crash 2026”); Maeil Shinmun, Oct 22, 2026; x.com/Jaemyung_Lee/status/2110145832511225984. 109 is Korea’s suicide-prevention line.

What “leverage” is, and why a 5% move can take everything

You put in $1,000. At 20× leverage you control a $20,000 position.



yours: $1,000 **borrowed from the exchange**: $19,000

price +5%

position worth $21,000



you doubled your money: +$1,000

price −5%

position worth $19,000: exactly the loan



you lost everything: −$1,000

the exchange closes your position automatically (**liquidation**); in a crash, millions of these happen at once and push the price further down

A 2× ETF (Korea) does this for you at 2×. On a crypto perpetuals exchange the multiplier is a **slider**, up to 50×.

The yield is visible on the way in. The risk, and who is on the other side, is not.

These products are moving toward US retail; Korea should be the lesson

In the US, right now

- **CLARITY Act** (crypto market structure): House 294 to 134, July 2025; one vote short in the Senate, Sept 15, 2026; revivable
- **CFTC**, Aug 2026: rules to let exchanges offer leveraged crypto trading to Americans; filed Sept 17
- **Hyperliquid** blocks US users today; its CEO is discussing a CFTC path

In Korea, after the crash

- new listings halted; minimum deposit tripled; exposure capped at 20% of assets. All of it *after* the losses.

On a public-wallet venue, every trade you make is on the tape

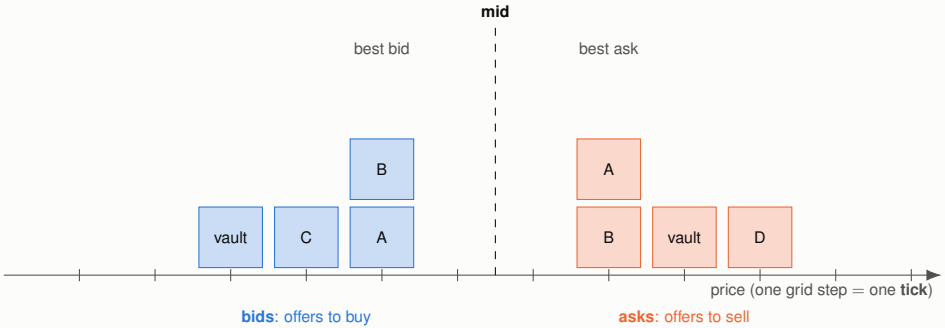
fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
public, forever; the professionals read it

What an everyday trader would need to know

- what a trade really costs
- who is on the other side
- whether their identity is used against them
- which venue rules protect them in stress

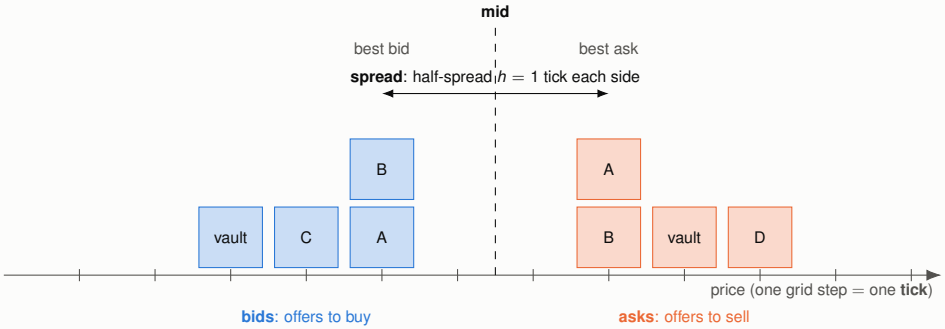
Korea's protections came after the losses. We measure these four things now, on the real venue and in a model where the rules can be changed, so the protections can come first.

An exchange is a list of standing offers: the order book



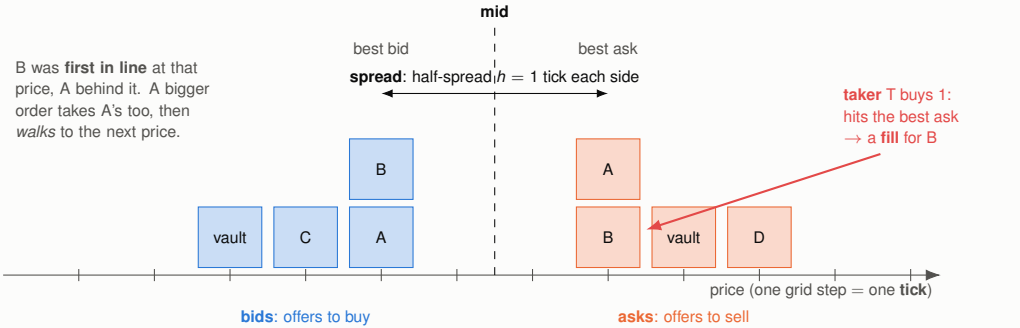
Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

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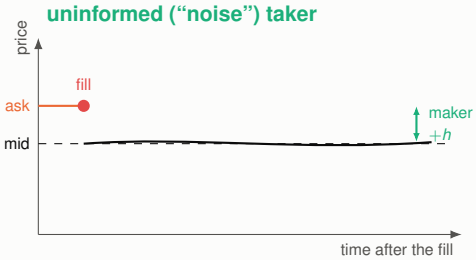
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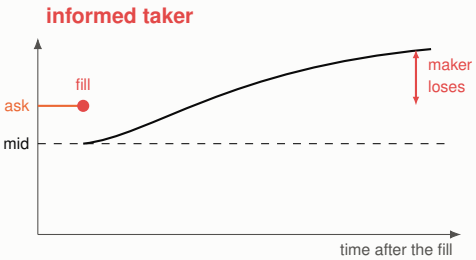
Makers post the offers (the boxes). **Takers** accept them. Each box carries a public **wallet** address.

Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread

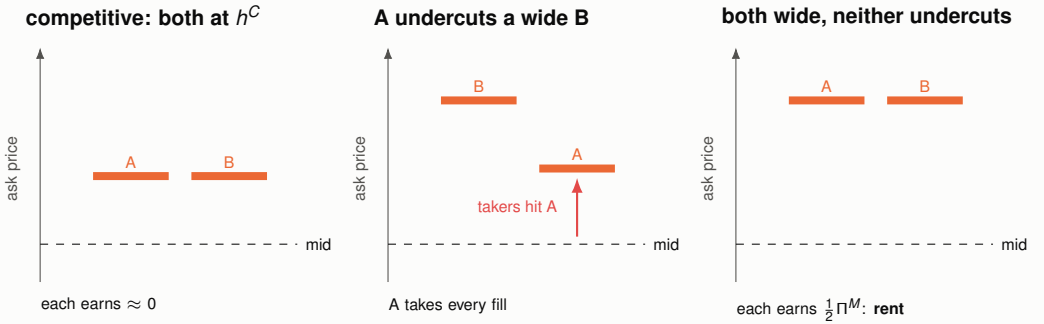


the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

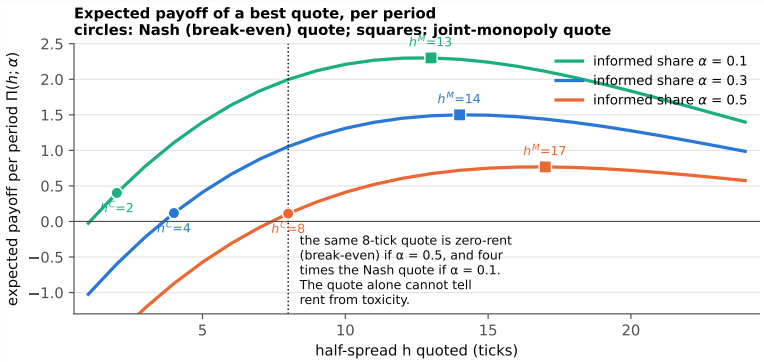
Rent: what makers earn by not undercutting each other



h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

The puzzle: on the tape, rent and toxicity look identical



A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Three players, and what each one gets to know

incumbent makers
learned a cartel; decide *how wide*, and whether to stay in line

an entrant
decides which quotes to undercut, which fills to be there for

the venue
decides the rules: public identities? tick size? pay for position?

the entrant's information set is the *treatment*



What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

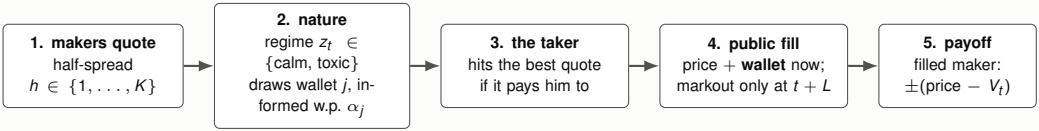
How much of a learned cartel's rent can an adversary take or destroy?

RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: the rent is fully exploitable; identity's value is a lag story; and on the real book the makers who earn are the ones avoiding the toxic wallets.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h; \alpha) =$
$(1 - \alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

Benchmarks are closed-form, so “rent” is measured, not estimated

$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\mathrm{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\mathrm{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

The prototype, in one command

`python experiments/demo.py`

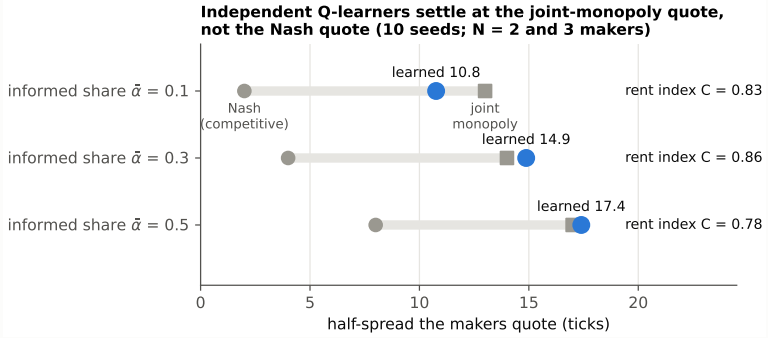
1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)              0.41           0.33      0.48
  markout (undercut if rent inferred)     0.47           0.41      0.55
  exact best response                     0.94           1.09      1.10

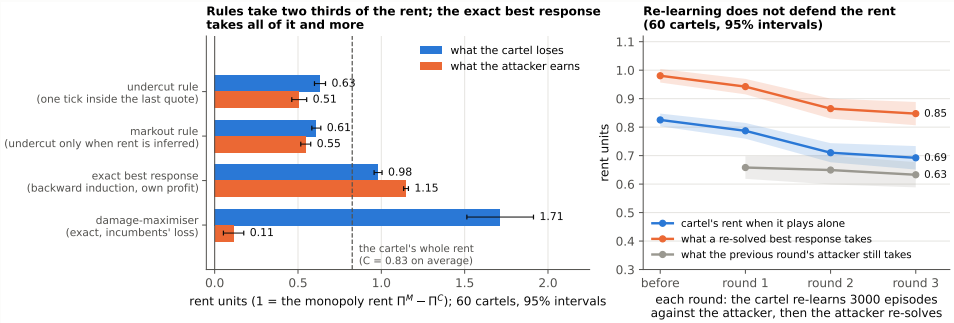
Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24       12   noise    buy  entrant   +1.21
   1    23    16       13   noise    buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed buy  entrant   -5.77
   7    18    23       14   noise    buy  entrant  +11.79
```

Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

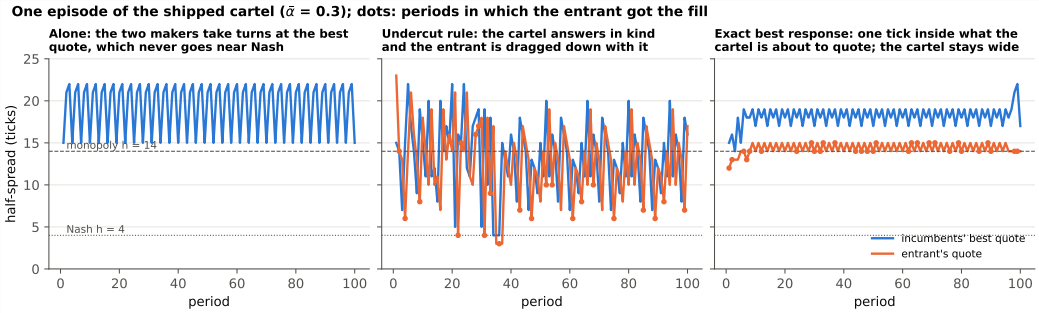
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3.

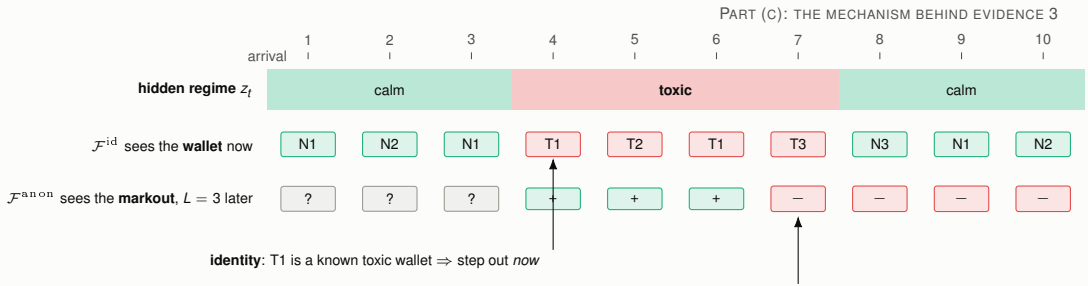
The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

Steering, period by period



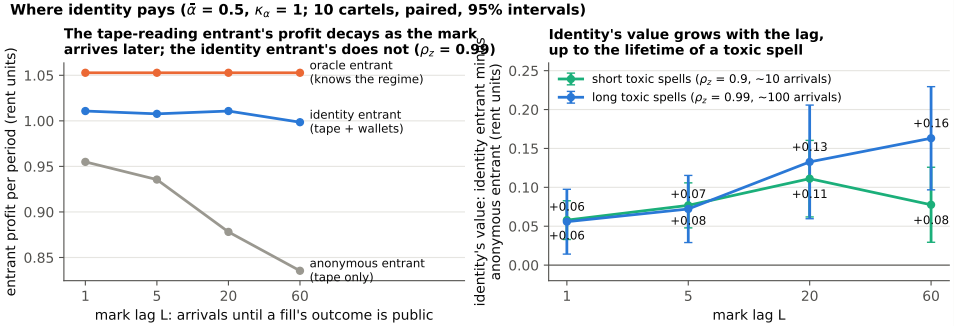
Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

Why identity has value: a spell of toxic flow, seen two ways



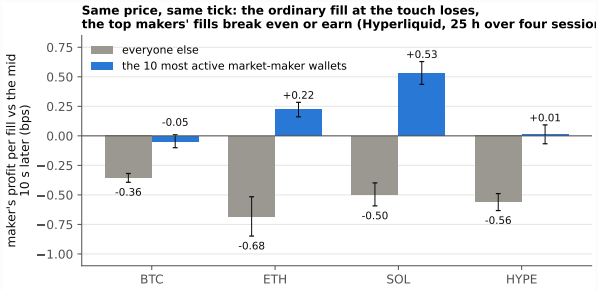
Identity is worth what you lose in the L fills before the tape catches up.

Evidence 3: what identity is worth, and why



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Evidence 4, Hyperliquid: the price cannot move, and someone still earns

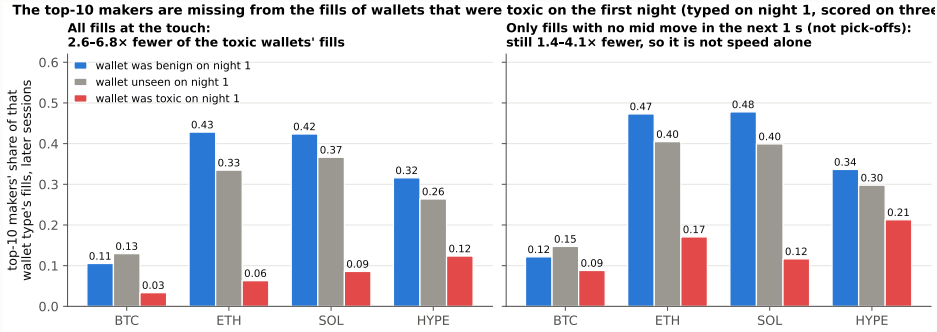


Four sessions, 25 h, 490k scored fills.

- **The spread is one tick**, essentially always
- **The ordinary fill loses**: -0.2 to -0.4 bps in 10 s
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

Rent lives in *which fills you get*, not in the quote.

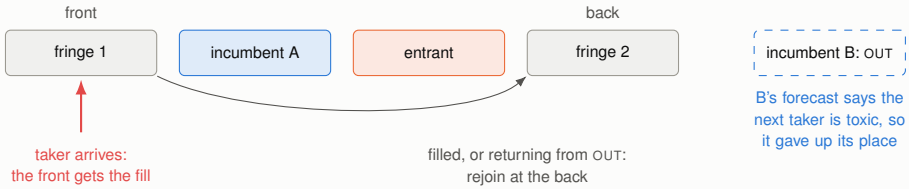
Evidence 5: how they do it is identity, not the tape



They never withdraw after a stressful spell (30 s to 5 min). They are absent from the toxic *wallets'* fills: typed on night 1, scored on three later sessions.

The queue game: the one-tick book

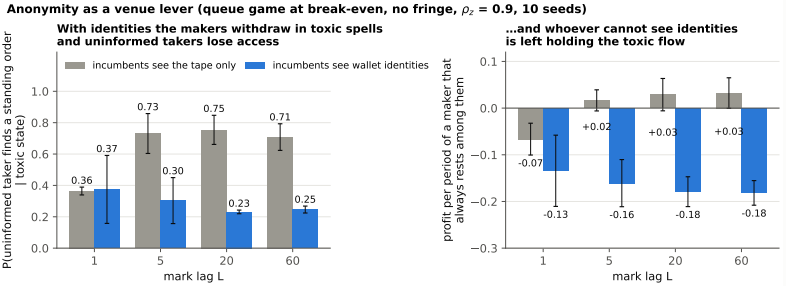
spread pinned at one tick; makers choose REST or OUT; the queue is first in, first out



A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays: whoever rests through the toxic spell, the *fringe* (retail orders, the vault).

Evidence 6: the levers decide who holds the toxic flow, not whether rent survives



Anonymity: a transfer from makers to uninformed takers in stress; identity the reverse.

Priority fee: the venue collects 2 to 4× the rent; the identity-blind fringe pays.

Makers' own levers: only an identity-aware floor helps, and it leaves the adversary 75 to 80%.

Plan for Workshop 2

Attacks and learning

1. **Public-information attacker**: identify the reaction function from the tape, then solve
2. **Both sides learning**: PSRO; are adversarially trained cartels less exploitable, or just less collusive?
3. **Queue-mode attacks** on no-regret incumbents

Designs, and data

- Venue and maker levers on one frontier: profit vs worst-5% under attack, plus takers' access in stress
- A month of recording, by coin, hour and regime; the order-level archive

Your view: is a frozen reaction function the right "realistic attacker"?

If you remember three things

1 Rent is not defensible

Solve the cartel's reaction function and you take the monopoly profit. Re-learning does not get it back.

2 Identity is a lag story

A wallet reveals a toxic spell before the price does. Worth zero when the tape is instant, more with the lag.

3 The real book agrees

On a one-tick book rent is which fills you get. The makers who earn avoid last night's toxic wallets.

`github.com/pradeepu191/wargame`

Recorder running continuously since October 25.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.